Frequency, Pitch, Timbre

Intro to Audio Tech I
Jocelyn Kavanagh
Wednesday, August 27

Georgia Tech

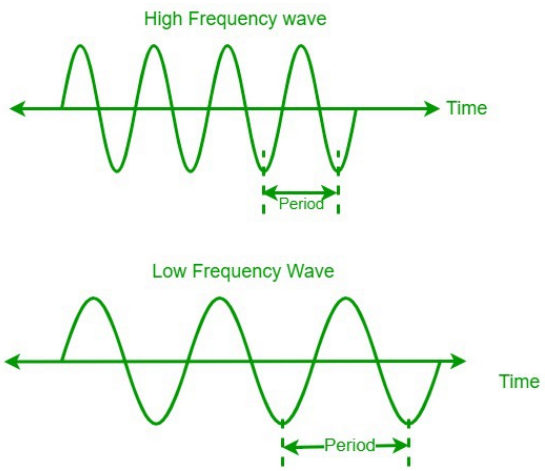
Lesson Outline

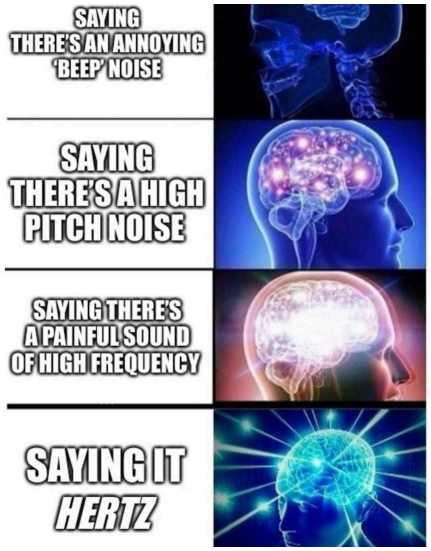
- Frequency and Pitch
- Resonance and Overtones
- Timbre
- Up Next…

Frequency

What is Frequency?

- The repetition of an “amplitude pattern/cycle” over time
- The number of cycles per second
- Measured in hertz (1/s)

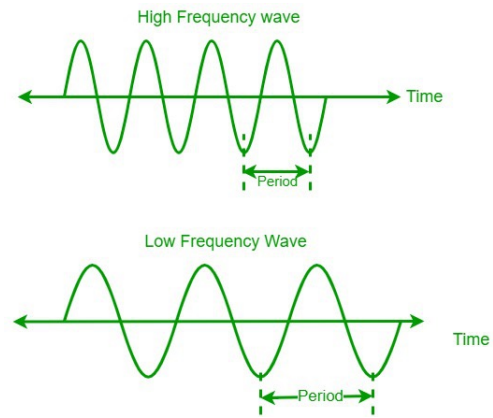




What is Period?

- Period
  - The amount of time it takes to complete a cycle
  - Inverse of frequency

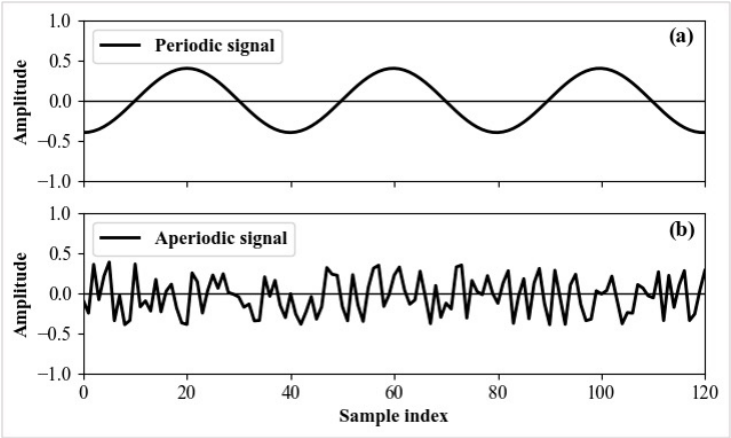
$$T = \frac{1}{f} = \frac{2\pi}{\omega} = \frac{360}{\omega}$$



- What is the period of a wave with an angular velocity, $\omega = 6$?

Periodic vs Aperiodic (reminder

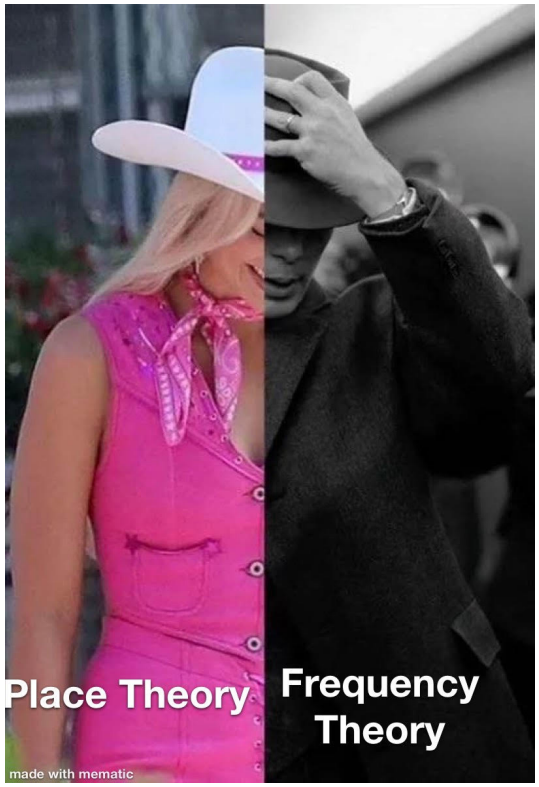
- A sound wave is mathematically periodic if it repeats a pattern infinitely
- We can consider something “musically periodic” if it repeats regularly enough to create the perception of pitch
- If a sound doesn’t follow a pattern, it is aperiodic or noise



Human Perception and Pitch

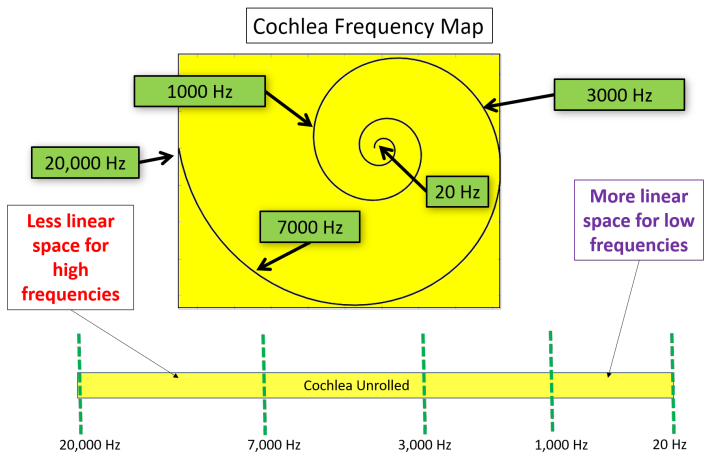
Detecting Frequency

- What's the range for human hearing?
  - 20 Hz to 20 kHz
- Frequency detection is roughly logarithmic
- Frequency has two theories…
  - Place Theory vs Frequency Theory
    - Place - where in the cochlea matters
    - Frequency - how many hairs/how fast the firing matters



Detecting Frequency

- Critical Bands
  - Relate to our ability to distinguish between individual frequencies
  - Better at low tones than high tones
- Roughness and Fluctuation

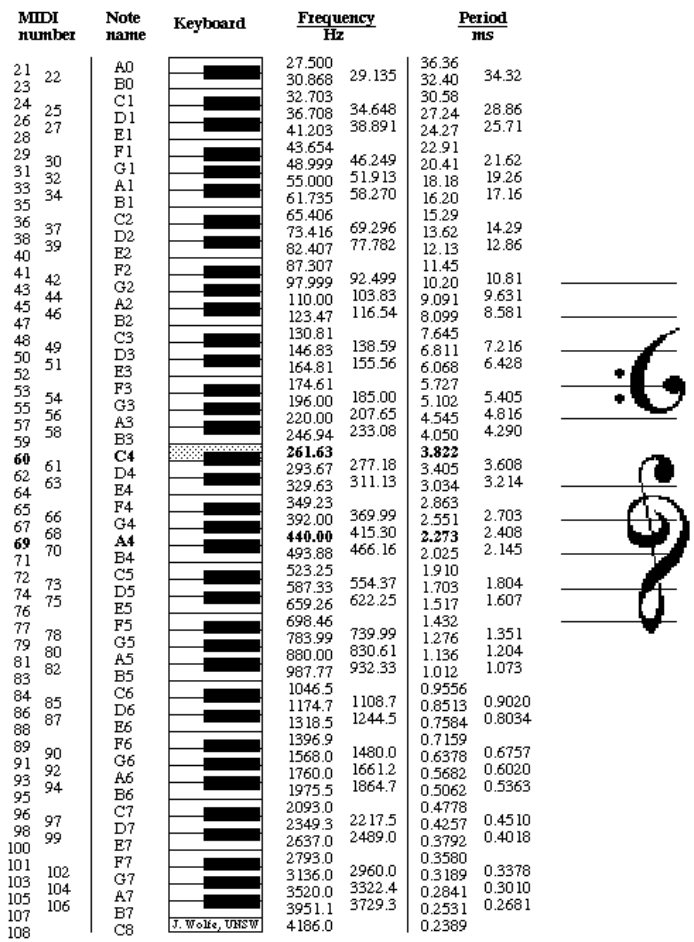


Critical Band (Bark)	Center Frequency (Hz)	Bandwidth (Hz)
1	50	100
2	150	100
3	250	100
4	350	100
5	450	110
6	570	120
7	700	140
8	840	150
9	1000	160
10	1170	190
11	1370	210
12	1600	240
13	1850	280
14	2150	320
15	2500	380
16	2900	450
17	3400	550
18	4000	700
19	4800	900
20	5800	1100
21	7000	1300
22	8500	1800
23	10500	2500
24	13500	3500

Georgia Tech

What is pitch?

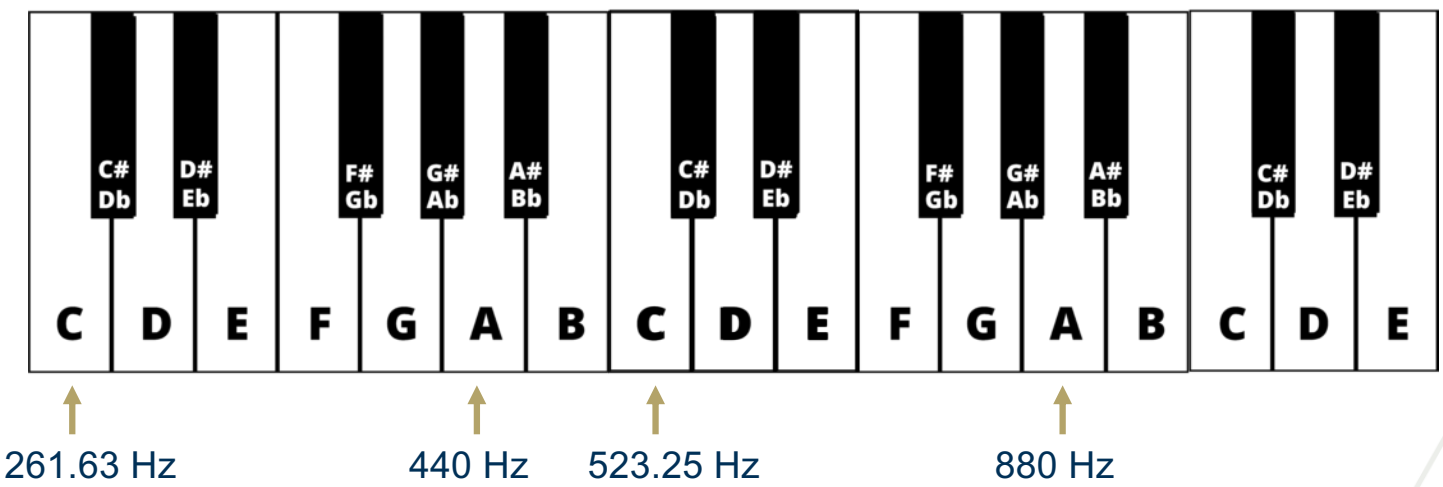
- Perception of frequency
- How ”high” or “low” a sound is
- Pitch is relative, while frequency is absolute
- Perceived as fixed ratios
  - Not fixed number of cycles per second



MIDI number	Note name	Frequency Hz	Period ms
21	A0	27.500	36.36
22	B0	29.135	34.32
23	C1	30.868	32.40
24	D1	32.703	30.58
25	E1	34.648	28.86
26	F1	36.708	27.24
27	G1	38.891	25.71
28	A1	41.203	24.27
29	B1	43.654	22.91
30	C2	46.249	21.62
31	D2	48.999	20.41
32	E2	51.913	19.26
33	F2	55.000	18.18
34	G2	58.270	17.16
35	A2	61.735	16.20
36	B2	65.406	15.29
37	C3	69.296	14.29
38	D3	73.416	13.62
39	E3	77.782	12.86
40	F3	82.407	12.13
41	G3	87.307	11.45
42	A3	92.499	10.81
43	B3	97.999	10.20
44	**C4**	103.83	9.631
45	D4	110.00	9.091
46	E4	116.54	8.581
47	F4	123.47	8.099
48	G4	130.81	7.645
49	**A4**	138.59	7.216
50	B4	146.83	6.811
51	C5	155.56	6.428
52	D5	164.81	6.068
53	E5	174.61	5.727
54	F5	185.00	5.405
55	G5	196.00	5.102
56	A5	207.65	4.816
57	B5	220.00	4.545
58	C6	233.08	4.290
59	D6	246.94	4.050
60	E6	**261.63**	**3.822**
61	F6	277.18	3.608
62	G6	293.67	3.405
63	A6	311.13	3.214
64	B6	329.63	3.034
65	C7	349.23	2.863
66	D7	369.99	2.703
67	E7	392.00	2.551
68	F7	415.30	2.408
69	G7	**440.00**	**2.273**
70	A7	466.16	2.145
71	B7	493.88	2.025
72	C8	523.25	1.910
73		554.37	1.804
74		587.33	1.703
75		622.25	1.607
76		659.26	1.517
77		698.46	1.432
78		739.99	1.351
79		783.99	1.276
80		830.61	1.204
81		880.00	1.136
82		932.33	1.073
83		987.77	1.012
84		1046.5	0.9556
85		1108.7	0.9020
86		1174.7	0.8513
87		1244.5	0.8034
88		1318.5	0.7584
89		1396.9	0.7159
90		1480.0	0.6757
91		1568.0	0.6378
92		1661.2	0.6020
93		1760.0	0.5682
94		1864.7	0.5363
95		1975.5	0.5062
96		2093.0	0.4778
97		2217.5	0.4510
98		2349.3	0.4257
99		2489.0	0.4018
100		2637.0	0.3792
101		2793.0	0.3580
102		2960.0	0.3378
103		3136.0	0.3189
104		3322.4	0.3010
105		3520.0	0.2841
106		3729.3	0.2681
107		3951.1	0.2531
108		4186.0	0.2389

Music Theory and Tuning

- Equally Tempered Chromatic Scale
  - Divide the octave in 12 equal tones (semitones)
  - Each semitone is equal to 100 cents
  - A semitone ratio is roughly 1.0595 : 1
- An octave is double the frequency (in Hz)
  - Interval not set frequency distance
  - C4 to C5 is an octave with ~261 Hz apart, while A4 to A5 is an octave with 440Hz apart



Complex Tones, Harmonics, Resonance

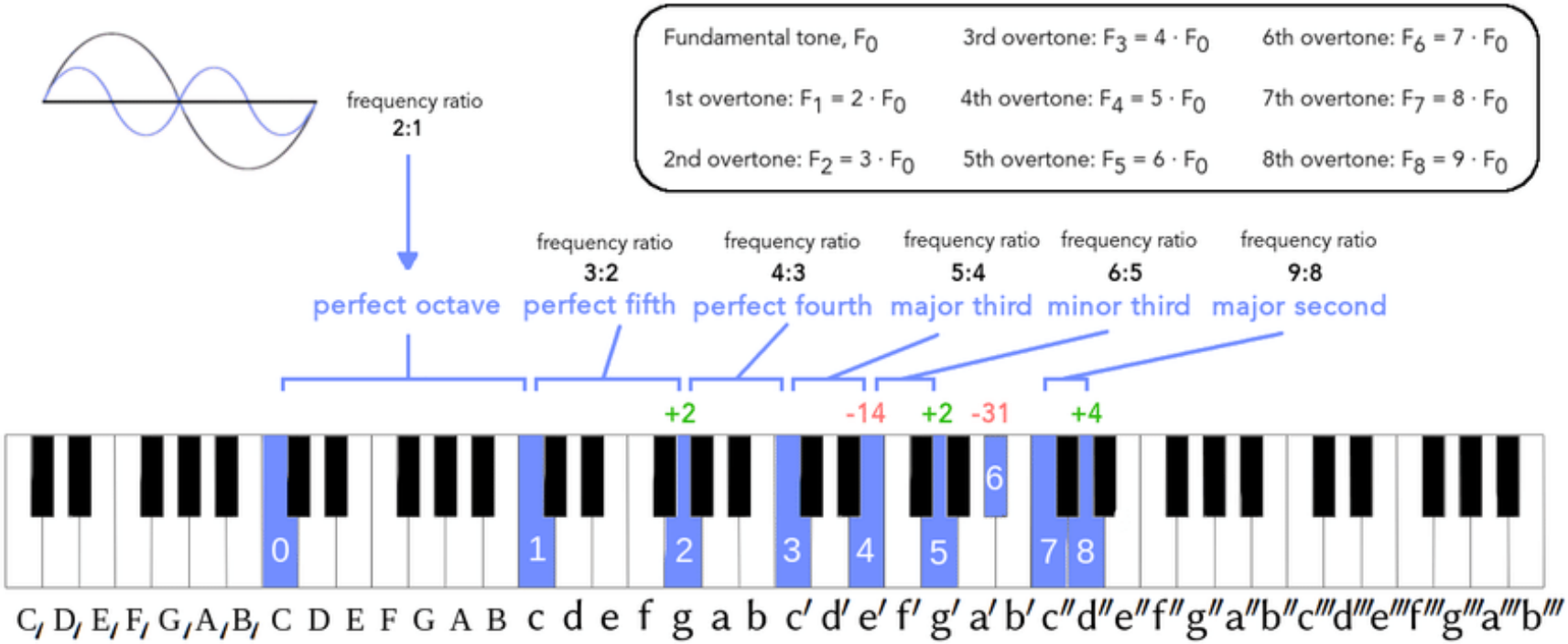
Harmonic Series

- Integer multiples of the fundamental frequency
- Partial -> single frequency within a complex sound
- Harmonic (partial) -> frequency with an integer relationship to the fundamental
- Inharmonic partial -> frequency without an integer relationship to the fundamental
- Also called overtones

Harmonic Number	Note Name	Ratio with Fundamental	Interval with Fundamental	Consecutive Ratio
12	G4	3 / 2	Perfect 5th	12 / 11
11	F♯4	11 / 8	Augmented 4th	11 / 10
10	E4	5 / 4	Major 3rd	10 / 9
9	D4	9 / 8	Major 2nd	9 / 8
8	C4	2 / 1	Octave	8 / 7
7	B♭3	7 / 4	Minor 7th	7 / 6
6	G3	3 / 2	Perfect 5th	6 / 5
5	E3	5 / 4	Major 3rd	5 / 4
4	C3	2 / 1	Octave	4 / 3
3	G2	3 / 2	Perfect 5th	3 / 2
2	C2	2 / 1	Octave	2 / 1
1	C1	1 / 1	Fundamental	

mu:zines

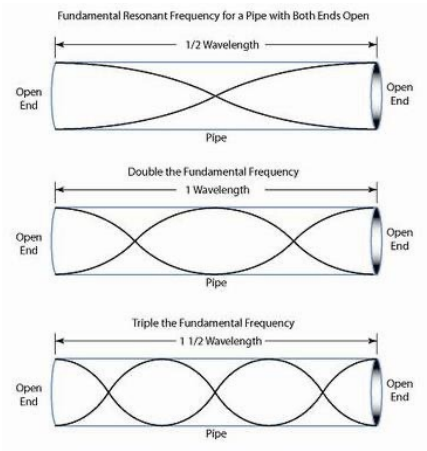
Harmonic Series

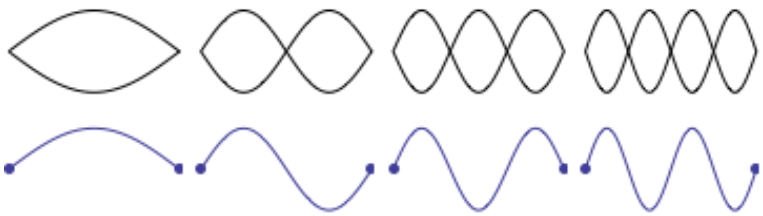


Georgia Tech

Resonance

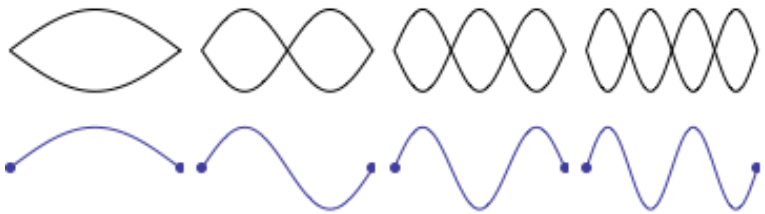
- Resonance occurs when matching vibrations from another object amplify the amplitude of an object's natural frequency
- Instruments -> strings, open columns, conical columns
- Objects have multiple modes of vibrations occurring at frequencies called harmonics
- The lowest resonant frequency is called the fundamental





Resonance and Harmonics

- When an object resonates, it creates a standing wave with large vibration
- This can only be done at certain frequencies, or harmonics



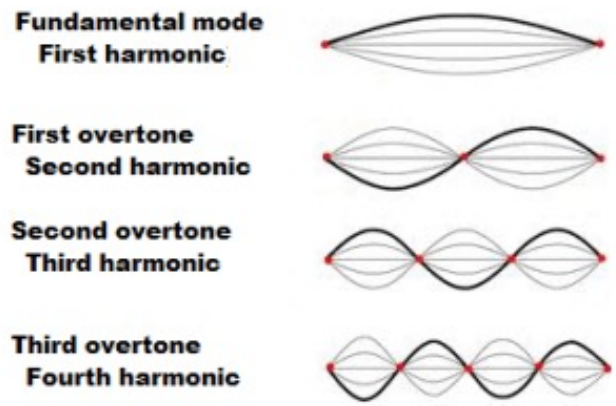
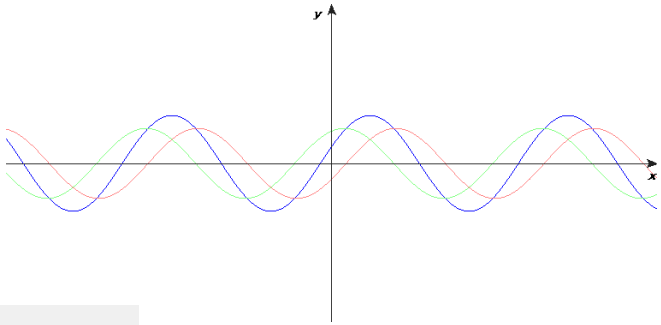
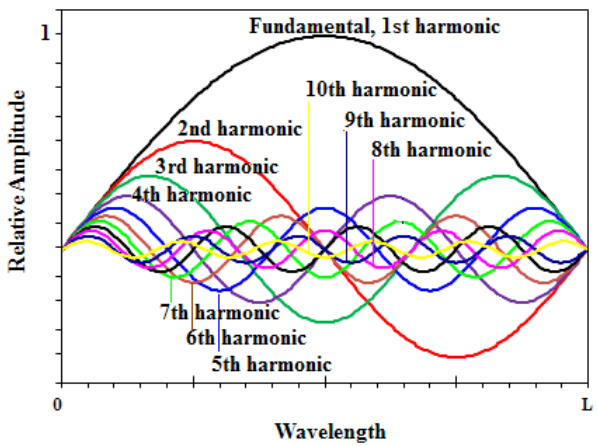


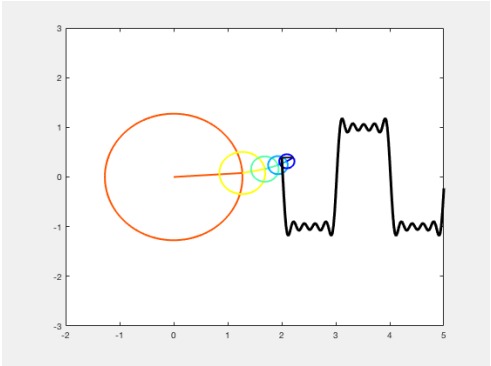
Fig. 1

Overtones Creating Complex Sounds

- When an object vibrates causing the fundamental, it may also vibrate along other modes of vibration causing overtones (or higher frequencies above f0)
- The cycling of the fundamental is the perceived pitch, but the harmonics will cycle many more times before the fundamental is complete
  - 1st harmonic will cycle twice
  - 2nd harmonic will cycle three times…

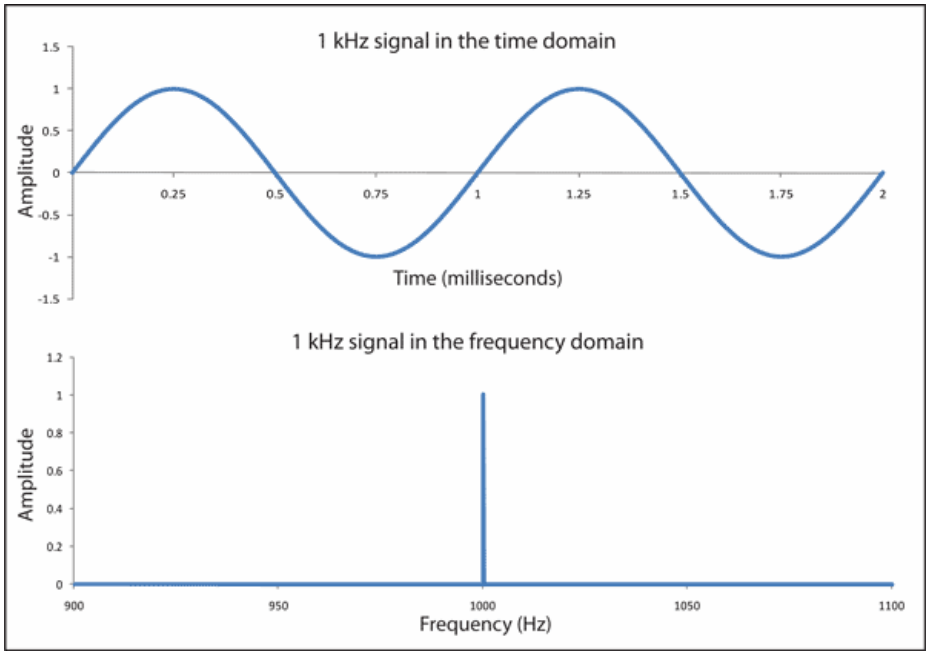






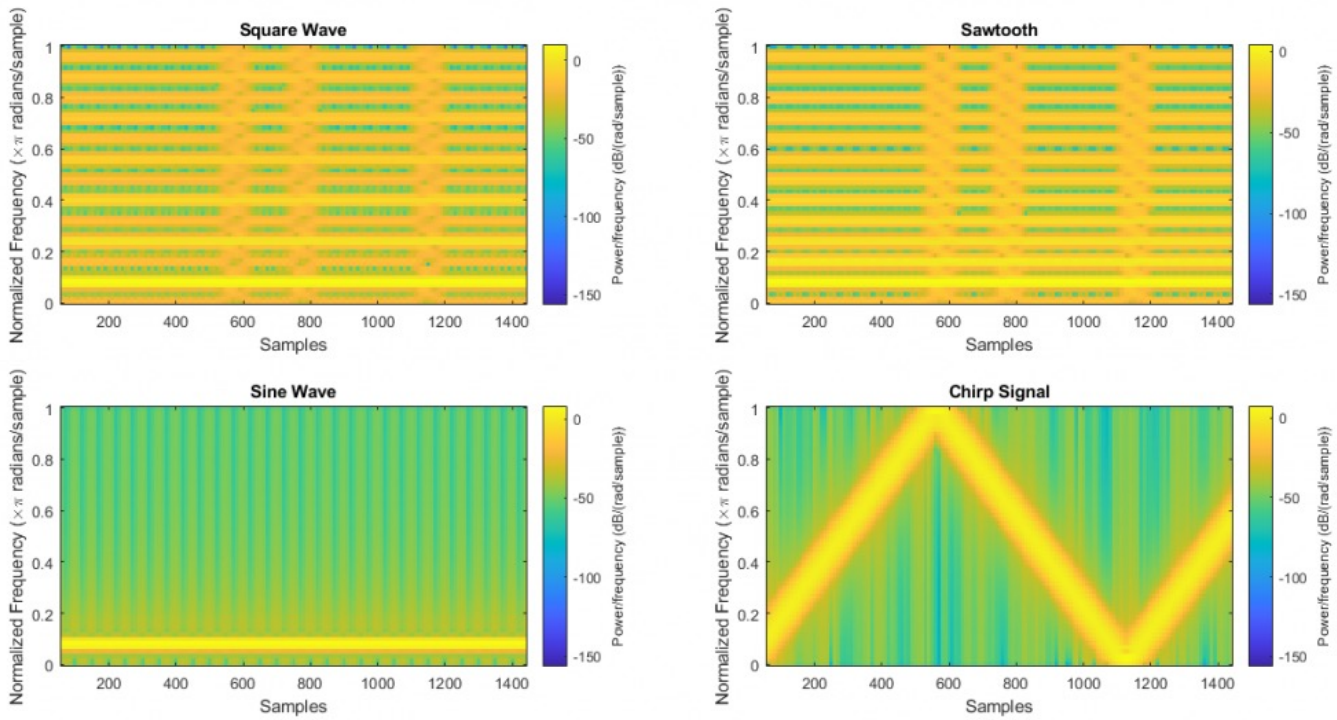
Step Back: Magnitude Spectrum

- The frequency spectrum tells us the frequencies present and their strengths
- Visualize the frequency components of a sound with a magnitude spectrum which shows frequency and magnitude/amplitude/power

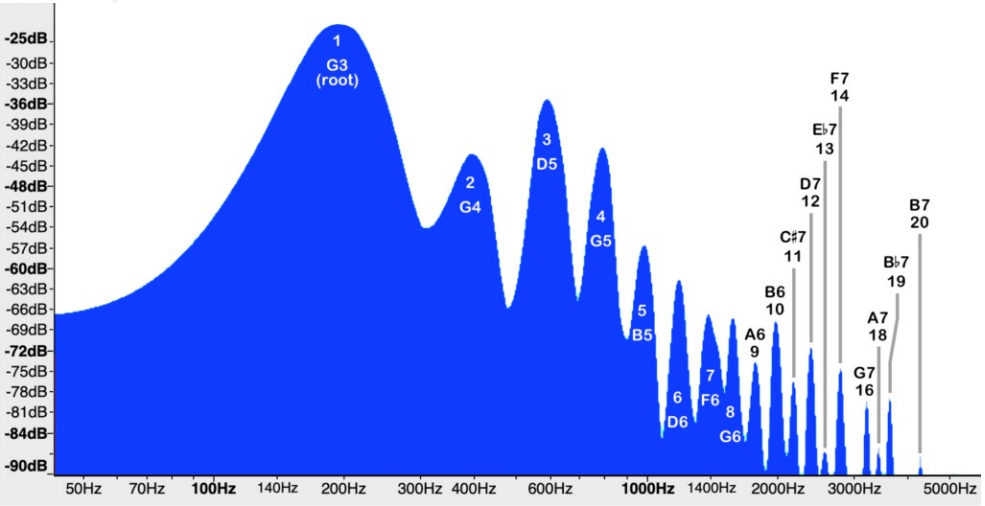


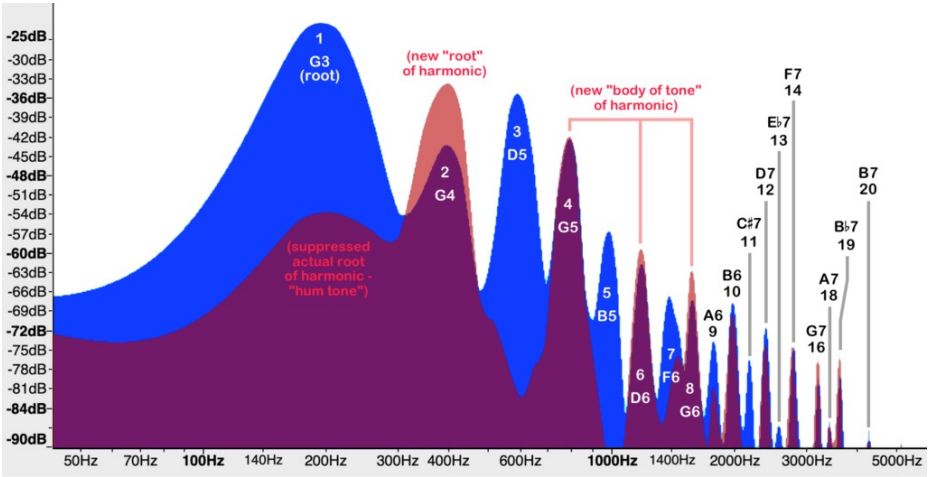
Step Back: Spectrogram

- We can also use a spectrogram which shows frequencies over time, with amplitude represented by color (3 dimensional)



Overtones of a Harp



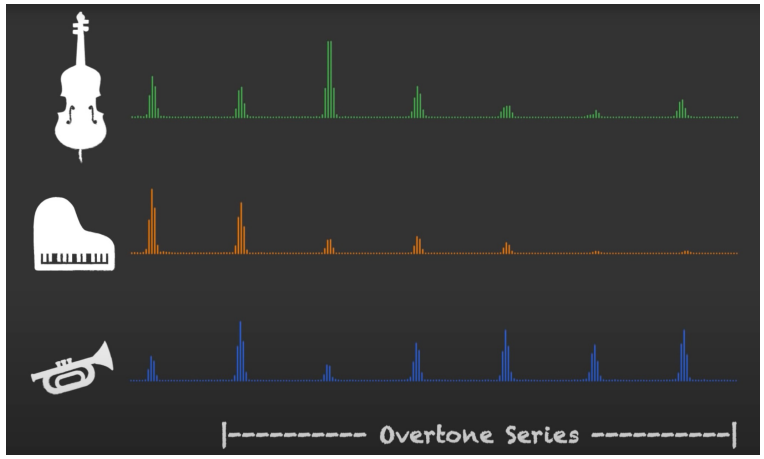


Georgia Tech

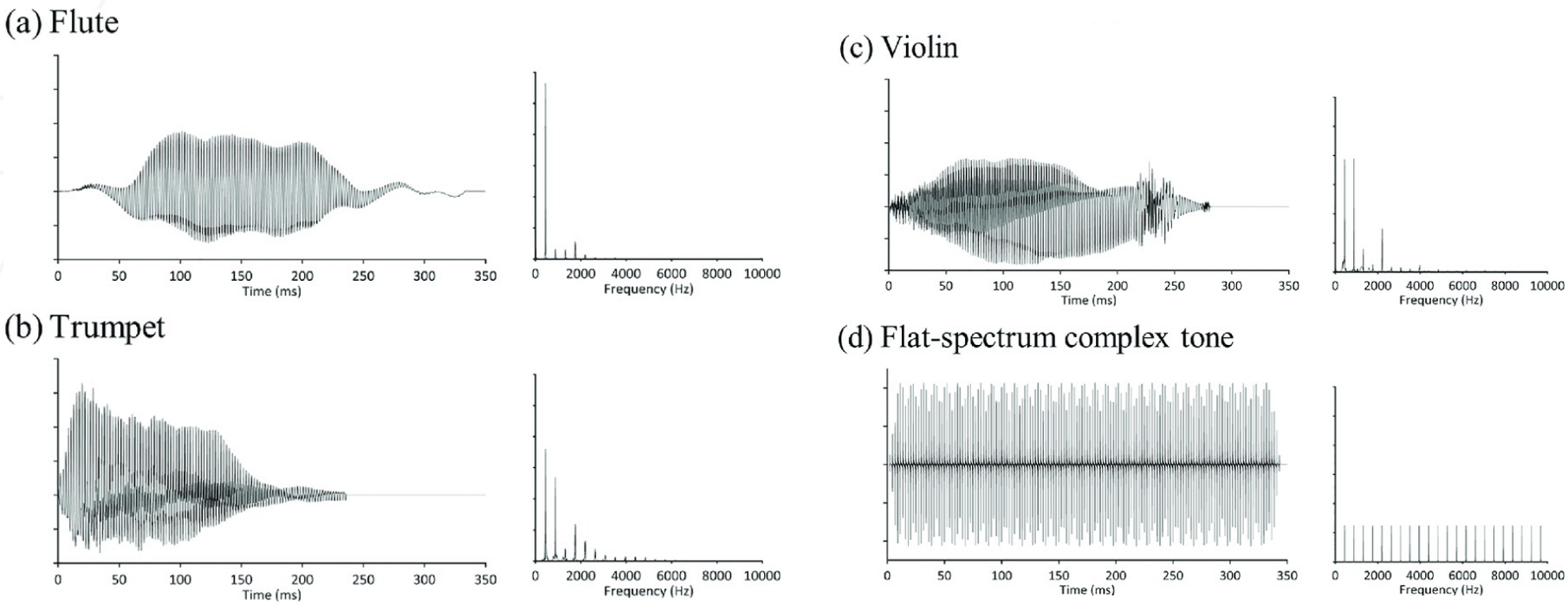
Timbre

What is timbre?

- Quality of the sound, “Tone Color”
- No great definitions…
  - ASA – “that attribute of auditory sensation which enables a listener to judge that two nonidentical sounds, similarly presented and having the same loudness and pitch, are dissimilar”
- Defined by the spectra and envelope of the sound



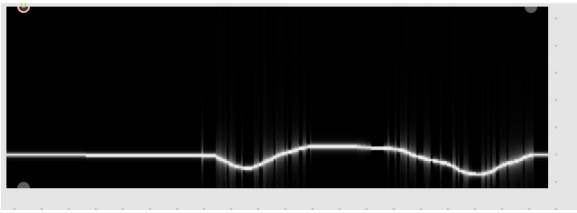
Spectrum - Overtones

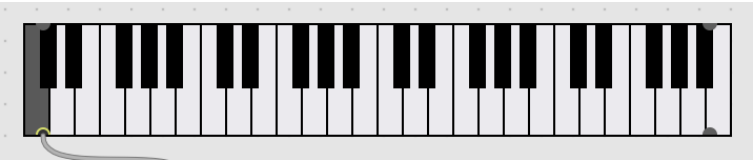


Frequency Control in Max

Frequency in Max

- Left inlet of cycle~ sets frequency
  - Can be a message, float, or a sequence of numbers (line, curve, etc)
- View frequency content with spectroscope (can also become a sonogram)
- Convert MIDI pitch to frequency using mtof
- Kslider helps you get MIDI





Up Next…

- Discussion post due on Friday
- Take-home quiz 1 will be posted on Friday and due the following Friday